Session 2: Practical Docker

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Course Materials

https://github.com/mcpp-volodymyr/docker-workshop

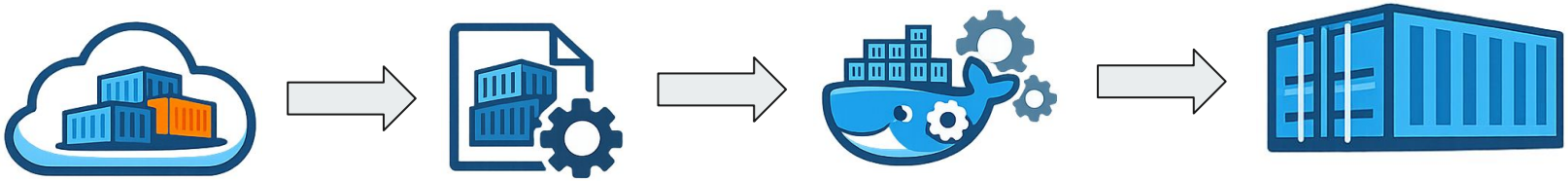
- Slides, examples, and commands fror this session are available in the repository
- No detailed note-taking required
- Focus on understanding the concepts and asking questions
- Start with README.md

Agenda

- Session 1 recap
- Homework and questions
- Running a web application

Docker Architecture Overview

- A **registry** stores images
- The **Docker engine** pulls **images** from a registry
- The engine creates and runs **containers** from those images
- A container is a running instance of an image



What Is a Dockerfile?

- A **Dockerfile** is a text file with instructions for building a Docker image
- *docker build* uses a Dockerfile to create an image

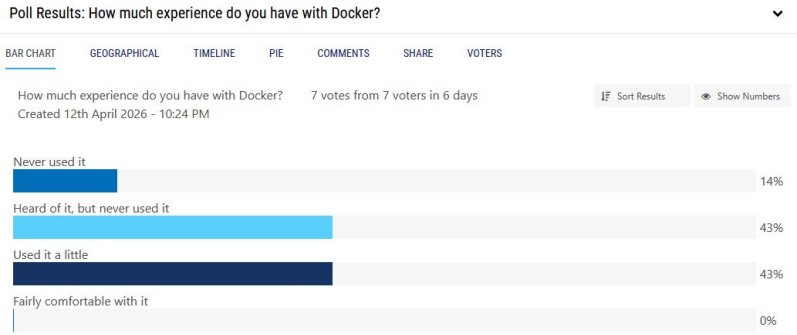
Build and Run the Image

docker build -t demo-http .

docker run --rm -p 8000:8000 demo-http

Web Poll

- Web app: displays the poll and results
- Database: stores the votes
- Goal: run the application with the web app connected to the database



Poll App Architecture

Dockerfile

```
FROM python:3.12-slim

ENV PYTHONUNBUFFERED=1

WORKDIR /app

COPY . .

RUN pip install --no-cache-dir -r requirements.txt

EXPOSE 8000

CMD ["python", "app.py"]
```

RUN

RUN pip install --no-cache-dir -r requirements.txt

- executes a command during image build

Build and Run the Web App

docker build -t webpoll .

docker run --rm -p 8000:8000 webpoll

What Is Docker Compose?

- *docker run* works well for a single container
- Real applications often need multiple services
- Example: web app + database
- Starting and configuring them manually becomes inconvenient
- Docker Compose lets us define and run them together

Docker Compose Main Workflow

- *docker compose up --build*
- *docker compose down*
- *docker compose ps*
- *docker compose logs -f*

Docker Compose Service-Specific Operations

- *docker compose up --build -d web*
- *docker compose restart db*

Useful Links

- https://github.com/mcpp-volodymyr/docker-workshop

  Session materials, slides, examples, and practice tasks

- https://docs.docker.com/reference/compose-file/

  Compose file reference

- https://docs.docker.com/reference/cli/docker/compose/

  Docker Compose CLI reference

Practice Task

- Follow the instructions in session-2/homework.md
- Use the repository materials as a reference
- Bring your questions to the next session

Thank You!